

- 4) Except rectangular data matrix with non-negative entries, its data requirement is highly flexible.

16.3 STEPS AND ALGORITHM INVOLVED IN CORRESPONDENCE ANALYSIS TECHNIQUE

CA technique involves the following steps and algorithm:

- 1) Construction of the primitive matrix.
- 2) Calculation of the row profile and the column profile.
- 3) Computation of the correspondence matrix.
- 4) Obtain total row mass & total column mass.
- 5) Construction of the augmented correspondence matrix. Sometimes for the sake of simplicity, it is also named as correspondence matrix.
- 6) Obtain the chi-square distance & inertia.
- 7) Reduction of the dimension of the data.
- 8) Obtain the biplots.

16.4 BASIC CONCEPTS AND DEFINITIONS

Let us understand the basic concepts and definitions of correspondence analysis with the help of an illustration. We are taking up a worked out example to understand some of the points involved in the correspondence analysis technique.

Let us consider 2 qualitative variables: Educational Status and Employment Status. Correspondence analysis technique will help us to examine the association between these two qualitative variables. For the purpose of our analysis, we have divided employment status into six categories: Employer, regular, pensioner/earning income from assets, temporary, contractual/own account worker and unemployed. Similarly, educational attainment of the persons have been categorized as illiterate, upto medium education, between medium and higher education, upto graduation, uptopost graduation or any technical degree and Ph.D and above.

Before examining the association between these two categorical variables, let us first understand the basic terms and definitions involved in the correspondence analysis technique.

16.4.1 Primitive Matrix

Primitive Matrix is the information which has been collected and classified in the tabular form. It is a cross tabulation of row and column variables. Table no. 16.1 offers a good example of Primitive Matrix. The information about employment status and educational attainment with their sub categories in rows and columns respectively has been provided in this table.

Table 16.1: Correspondence Between Educational Attainment and Employment Status**Primitive Matrix ($A = a_{ij}$)**

Educational Attainment	Employment_Status						
	unemployed	Contractual	Temporary	Pensioner	Regular	Employer	Active Margin
Illiterate	22	16	1	1	0	1	41
Upto Middle	27	7	0	1	7	2	44
Middle to Higher Secondary	69	8	3	2	4	12	98
Graduation	51	2	4	4	22	20	103
Post grad/Technical degree	13	11	7	0	46	14	91
PhD & Above	1	3	4	0	28	3	39
Active Margin	183	47	19	8	107	52	416

Source: Assessment of Human Wellbeing: A Multidimensional Approach, a research study (unpublished) undertaken in 2015 by Prof. Narayan Prasad.

16.4.2 Profiles

Profiles display the characteristics of a particular category which is determined by their respective frequencies. As shown in the table no. 16.2a, row profile describes the educational status while the column profile displays the employment status.

Row profiles give the frequency of each row point such that summation for row will be equal to 1. Row profile are calculated by taking individual row point divided by their corresponding row total (active margin). That is,

$$\begin{aligned} \text{row profile for the } i^{\text{th}} \text{ row} &= \frac{\text{every entry of the } i^{\text{th}} \text{ row of } A}{\text{sum of the entries of the } i^{\text{th}} \text{ row of } A} \\ &= \frac{a_{ij}}{\sum_{j=1}^n a_{ij}} \end{aligned}$$

Column profiles are calculated in similar manner.

$$\begin{aligned} \text{Column profile for the } j^{\text{th}} \text{ column} &= \frac{\text{every entry of the } j^{\text{th}} \text{ column of } A}{\text{sum of the entries of the } j^{\text{th}} \text{ column of } A} \\ &= \frac{a_{ij}}{\sum_{i=1}^m a_{ij}} \end{aligned}$$

The only difference is that in this case, each column point is divided by their corresponding column total (active margin) as shown in table 16.2b.

Table 16.2a: Correspondence Between Educational Attainment and Employment Status**Row Profiles**

Educational Attainment	Employment_Status						
	unemployed	Contractual	Temporary	Pensioner	Regular	Employer	Active Margin
Illiterate	.537	.390	.024	.024	.000	.024	1.000
Upto Middle	.614	.159	.000	.023	.159	.045	1.000
Middle to Higher Secondary	.704	.082	.031	.020	.041	.122	1.000
Graduation	.495	.019	.039	.039	.214	.194	1.000
Post grad/Technical degree	.143	.121	.077	.000	.505	.154	1.000
PhD & Above	.026	.077	.103	.000	.718	.077	1.000
Mass	.440	.113	.046	.019	.257	.125	

Table 16.2b: Correspondence Between Educational Attainment and Employment Status**Column Profiles**

Educational Attainment	Employment_Status						
	unemployed	Contractual	Temporary	Pensioner	Regular	Employer	Mass
Illiterate	.120	.340	.053	.125	.000	.019	.099
Upto Middle	.148	.149	.000	.125	.065	.038	.106
Middle to Higher Secondary	.377	.170	.158	.250	.037	.231	.236
Graduation	.279	.043	.211	.500	.206	.385	.248
Post grad/Technical degree	.071	.234	.368	.000	.430	.269	.219
PhD & Above	.005	.064	.211	.000	.262	.058	.094
Active Margin	1.000	1.000	1.000	1.000	1.000	1.000	

16.4.3 Masses

The mass of the i^{th} row is the marginal frequency of i^{th} row divided by grand total. Marginal frequency of the i^{th} is the sum of the entries of the i^{th} row. Mass for each row and column points are depicted in table no. 16.2a & 16.2b respectively. Grand total in this particular illustration is total sample size which is 416. Similarly mass for j^{th} column is marginal frequency of j^{th} column divided by grand total.

16.4.4 Correspondence Matrix

Correspondence matrix is defined as the original matrix divided by grand total. The correspondence matrix shows how value of mass is distributed across the cells. The row total and the column total of the correspondence matrix are the row mass and column mass respectively as shown in table 16.3.

Table 16.3: Correspondence Between Educational Attainment and Employment Status**Correspondence Matrix**

Educational Attainment	Employment_Status					
	unemployed	Contractual	Temporary	Pensioner	Regular	Employer
Illiterate	0.053	0.038	0.002	0.002	0.000	0.002
Upto Middle	0.065	0.017	0.000	0.002	0.017	0.005
Middle to Higher Secondary	0.166	0.019	0.007	0.005	0.010	0.029
Graduation	0.123	0.005	0.010	0.010	0.053	0.048
Post grad/Technical degree	0.031	0.026	0.017	0.000	0.111	0.034
PhD & Above	0.002	0.007	0.010	0.000	0.067	0.007

16.4.5 Augmented Correspondence Matrix

Having computed the correspondence matrix, we now append this correspondence matrix by adding the column of row masses and the row of column masses in the table 16.4a.

Table 16.4 a: Correspondence Between Educational Attainment and Employment Status**Augmented Correspondence Matrix**

Educational Attainment	Employment_Status						Row Mass
	unemployed	Contractual	Temporary	Pensioner	Regular	Employer	
Illiterate	0.053	0.038	0.002	0.002	0.000	0.002	0.099
Upto Middle	0.065	0.017	0.000	0.002	0.017	0.005	0.106
Middle to Higher Secondary	0.166	0.019	0.007	0.005	0.010	0.029	0.236
Graduation	0.123	0.005	0.010	0.010	0.053	0.048	0.248
Post grad/Technical degree	0.031	0.026	0.017	0.000	0.111	0.034	0.219
PhD & Above	0.002	0.007	0.010	0.000	0.067	0.007	0.094
Column Mass	0.440	0.113	0.046	0.019	0.257	0.125	1.000

16.4.6 Inertia

The term inertia, in the correspondence analysis explains the degree of variation. The value of inertia is high when there is large deviation of row and column profiles from their averages. The method to compute the total inertia is as follows:

$$\text{Total inertia} = \sum_i \sum_j \frac{(p_{ij} - r_i c_j)^2}{r_i c_j}$$

Where p_{ij} = (i, j)th entry of correspondence matrix

R_i = i th row profile

C_j = j th column profile

The inertia in table no. 16.3 gives the total variance explained by each dimension in the model. In this illustration value of inertia is 0.480. This value of inertia indicates that in this model knowing something about education attainment explains around 48% of something about employment status and vice versa. This association is highly significant as indicated by chi square statistic.

16.4.7 Distance

Independence and association between the two variables is measured by chi square. Chi square distances are weighted Euclidean distance between the profile points. Weights here refers to weights of dimension and not to the weights of profile points. Chi square is calculated as follows:

Formula to compute the chi-square distance:

Suppose the correspondence matrix of the primitive matrix is $m \times n$. Then, we define:

Chi-square distance between r^{th} and s^{th} rows is:=

$$d(r, s) = \sqrt{\sum_{j=1}^n \frac{1}{a_j} \left(\frac{a_{rj}}{a_r} - \frac{a_{sj}}{a_s} \right)^2}$$

where a_j = marginal frequency of j^{th} column

$$= \sum_{i=1}^m a_{ij}$$

a_r = marginal frequency of r^{th} row

$$= \sum_{j=1}^n a_{rj}$$

a_s = marginal frequency of s^{th} row

$$= \sum_{j=1}^n a_{sj}$$

16.5 REDUCTION OF DIMENSIONALITY

Another way of looking at correspondence analysis is to consider it as a method for decomposing the overall inertia by identifying a small number of dimensions in which the deviations from the expected values can be represented. This is similar to the goal of factor analysis, where the total variance is decomposed, so as to arrive at lower – dimensional representation of variables that allows one to reconstruct most of the variance/covariance matrix of variables.

Notice the tables 16.4 given below. The educational attainment column shows only 5 categories. Also notice that employment status has been condensed into two categories only – employed and unemployed. This reduction of dimensionalities is not arbitrary. It is a result of classification of sub categories in distances from same point of reference. This will further help in pictorial presentation of correspondence between the two variables – namely education attainment and employment status. Correspondence analysis aims to decompose the overall inertia so as to arrive at a lower dimensional space. Only those dimensions are included in the model which can easily be

interpreted. That is why the value of inertia does not always add up to 1 as all dimensions are not included in the model.

Table 16.4: Correspondence Between Educational Attainment and Employment Status

Summary

Dimension	Singular Value	Inertia	Chi Square	Sig.	Proportion of Inertia		Confidence Singular Value	
					Accounted for	Cumulative	Standard Deviation	Correlation
								2
1	.594	.353			.735	.735	.035	.009
2	.329	.108			.226	.960	.054	
3	.113	.013			.026	.987		
4	.069	.005			.010	.996		
5	.041	.002			.004	1.000		
Total		.480	199.660	.000 ^a	1.000	1.000		

a. 25 degrees of freedom

16.6 BILOTS

The biplot is concerned with data reconstruction in a joint map of the rows and columns, rather than distance reconstruction. In the simple case of a two-way table one can think of reconstructing different variants of the table depending on the way we think of the table: either as a set of rows, or a set of columns, or just a two-way table of entries where rows and columns are symmetric entities.

A biplot is a graphical representation of the information in an $n \times p$ data matrix. The bi- refers to the two kinds of information contained in a data matrix. The information in the rows pertains to samples or sampling units and that in the columns pertains to variables. When there are only two variables, scatter plots can represent the information on both the sampling units and the variables in a single diagram. This permits the visual inspection of the position of one sampling unit relative to another and the relative importance of each of the two variables to the position of any unit.

With several variables, one can construct a matrix array of scatter plots, but there is no one single plot of the sampling units. On the other hand, a two-dimensional plot of the sampling units can be obtained by graphing the first two principal components. The idea behind biplots is to add the information about the variables to the principal component graph.

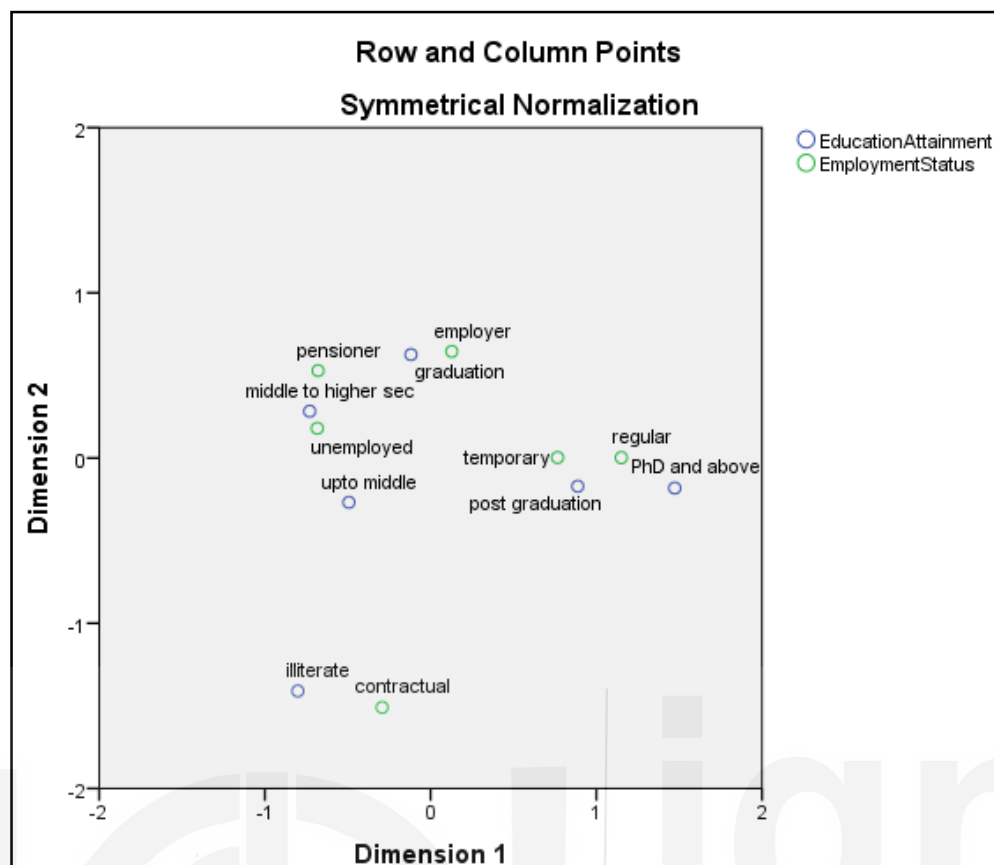


Fig.16.1: Biplots showing correspondence between Educational Attainment and Employment Status

Data has been graphically represented by the biplot in two dimensional space as shown in figure 1. However, there is one caution in this particular model that this model explains only 48% of employment status based on educational attainment. From the above biplot some general trends can be seen. Two points closer to each other shows greater degree of association between them as compared to others. This plot is offering us significant insight into direction of correspondence between the level of educational attainment and employment status. For example, we see that illiterate individuals are more likely to be employed on the contractual basis or they are mostly engaged in low level of own account work. Those individuals who have studied upto middle school are mostly found to be unemployed. Those with Ph.D and above education has higher possibilities of regular employment. The status of employment can be seen improving for individuals who have completed graduation and post-graduation as they are one who provides employment opportunities to others also. Therefore, this plot clearly reflects a greater degree of association between educational attainment and employment status which is also significant as confirmed by the value of chi-square.

16.7 INTERPRETATION OF THE RESULTS OF CORRESPONDENCE ANALYSIS

The interpretation of the results of correspondence analysis comprises the interpretation of numerical results and factor graphics yielded by CA. The former implies selection of significant axes and significant points.

We are taking up a worked out example for Correspondence Analysis to illustrate some of the points involved in the interpretation of the results of the correspondence analysis. Table 16.1 has information on 416 persons, their educational status and their employment status. The data originated from the study entitled 'Assessment of Human Wellbeing: A Multidimensional Approach' based on a sample survey conducted in Delhi in 2015. Through this technique, an attempt was made to find out correspondence between educational attainment and employment status among the respondents selected for the study. Educational status is subdivided into 6 categories, ranging from illiterate to Ph.D and above. The employment status varies from outright unemployed to contractual, temporary, regular, pensioner and also where respondents are employers rather than employees.

The table 16.4 shows summary of our calculation. The first column shows the five dimension – notice that as we are looking at distance from some dimension – the number of dimension declines from 6 to 5. The singular value and inertia are reported in column 2 to 3 respectively. Column 6 shows proportion of inertia accounted for by the first dimension which is highest at 0.735 of the total. As we move through dimension 2 onwards, inertia falls rather sharply. Notice the similarity with principal components analysis – there too, the first component explained largest variation. The column 7 shows cumulative proportion of inertia accounted for.

The table 16.5 gives an overview of row points. Column 2 shows mass & column 3 & 4 respectively show scores in dimension 1 & 2. Column 5 records inertia of respective categories in column 1. Column 6 & 7 record contribution of a point to inertia of dimension 1 & 2 respectively, while column 8 & 9 describe contribution of two dimension to inertia of the point concerned last column show total inertia.

Table 16.5: Correspondence Between Educational Attainment and Employment Status

Overview Row Points^a

EducationAttainment	Mass	Score in Dimension		Inertia	Contribution				
		1	2		Of Point to Inertia of Dimension		Of Dimension to Inertia of Point		
					1	2	1	2	Total
illiterate	.099	- 1.534	.948	.179	.350	.321	.860	.137	.997
upto middle	.106	-.771	.275	.046	.095	.029	.912	.048	.960
middle to higher secondary	.236	-.458	- .503	.050	.074	.216	.654	.328	.982
graduation	.248	.108	- .511	.022	.004	.235	.085	.798	.883
post graduation	.219	.975	.295	.146	.313	.069	.941	.036	.977
PhD and above	.094	1.074	.619	.085	.163	.130	.840	.116	.956
Active Total	1.000			.528	1.000	1.000			

The table 16.6 sums up similar information for column points.

Table 16.6

Overview Column Points^a

EmploymentStatus	Mass	Score in Dimension		Inertia	Contribution				
		1	2		Of Point to Inertia of Dimension		Of Dimension to Inertia of Point		
					1	2	1	2	Total
unemployed	.440	-.684	.180	.129	.346	.043	.948	.036	.984
contractual	.113	-.292	- 1.510	.091	.016	.782	.063	.926	.989
temporary	.046	.766	.003	.019	.045	.000	.858	.000	.858
pensioner	.019	-.679	.529	.011	.015	.016	.472	.158	.630
regular	.257	1.151	.002	.204	.574	.000	.992	.000	.992
employer	.125	.128	.645	.026	.003	.158	.046	.658	.705
Active Total	1.000			.480	1.000	1.000			

a. Symmetrical normalization

The figure 16.1 presents symmetrical normalization of the row and the column points in form of a picture in two dimension plane.

We must note that in our table 16.2a as we move up the educational attainment ladder, initially unemployment proportion is rising from 53.7% for illiterates to 61.4% for middle school educated and then 70% for middle to higher secondary level of education. For graduates, the proportion of unemployed come down to 49.5%. For post graduates, it is merely 14.3% and for highest educational category, it is barely 2.6%. Even, this data alone is offering us significant insight into direction of correspondence between the levels of educational attainments and employment status. The sub-categories of employment – contractual, temporary, regular, employer are shown separately and the data further tends to support the hypothesis that higher level of education attainment leads to higher possibilities of regular employment. However, the sub-category of ‘graduates’ also performs credibly as employers, while Ph.D and above get more regular employment.

Our correspondence analysis tends to provide us ‘concrete’ evidence in support of our tentative ‘conclusion’ from inspection of data described above.

16.8 MULTIPLE CORRESPONDENCE ANALYSIS

In a two-way contingency table, simple correspondence analysis provides association between two categorical variables. On the other hand, multiple correspondence analysis, an extension of the simple correspondence analysis, tackles the more general problem of associations among a set of more than two categorical variables. Generalization to more than two variables is neither obvious nor well-defined. In other areas of multivariate analysis, such as regression and log linear modelling, the situation is less complicated: for example, the transition from the regression of a response variable on a single predictor variable to the case of several predictors is quite straightforward.

The main problem is that the notion of association between two categorical variables is already a complex concept and there are several ways to generalize this concept to more than two variables.

Many different approaches exist in literature to define multiple correspondence analysis. Important among them are two main approaches: first, the correlation between sets of variables, known as canonical correlation, and second, the geometric approach, which is directly linked to data visualization, and which has many similarities to Pearson-style principal component analysis. In the explanation of each approach, one can consider the case of two variables only and then describe possible generalizations to more than two variables.

Check Your Progress 1

- 1) What is correspondence matrix?

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- 2) A sample of $n = 416$ people is given in the following table showing the annual income and level of education:

Table: Income level and level of educational attainment

Educational Attainment	Annual Income				
	Upto 2.50 Lakh	2.50 lakh to 5.00 lakh	5.00 lakh to 10.00 lakh	10.00 lakh to 20.00 lakh	Above 20.00 lakh
Illiterate	40	1	0	0	0
Up to Middle	28	3	8	3	2
Middle to higher secondary	44	18	23	11	2
Graduation	26	18	25	27	7
Post Graduation	2	6	25	31	27
Ph.D& Above	1	2	6	18	12

Compute the value of chi-square statistics and inertia from the above table. Also interpret the results.

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